400-Tonne Hydraulic Press (HYD-PRESS) Equipment Manual and SOP

VINDHYA PRECISION WORKS ## MAINTENANCE ENGINEERING DIVISION ### STANDARD OPERATING PROCEDURE: ASSET CLASS HYD-PRESS Document ID: SOP-HYD-400T-04 Applies To: 400-Tonne Hydraulic Press (Assets HYD-01 through HYD-06) Effective Date: 14 January 2025

1. PURPOSE AND SCOPE This document defines the mandatory operating limits, condition-monitoring thresholds, preventive maintenance schedules, fault isolation procedures, torque specifications, and consumable part numbers for the 400-Tonne Hydraulic Press (Asset Class HYD-PRESS). Compliance with this procedure is mandatory for all maintenance and operations personnel at Vindhya Precision Works.

2. NORMAL OPERATING RANGES AND MONITORING THRESHOLDS

2.1 Parameter Specifications The 400-Tonne Hydraulic Press shall operate strictly within the parameters defined in Table 2.1. The condition-monitoring system (CMS) is programmed with these exact numeric limits.

Table 2.1: Operating Ranges and Alarm Setpoints for HYD-PRESS

Parameter	Healthy Operating Range	Warning Setpoint	Trip / Interlock Setpoint	Corrective Action Trigger
Vibration (RMS)	1.2 – 4.0 mm/s	5.6 mm/s RMS	8.5 mm/s RMS	Immediate shutdown at trip
Surface Temperature	38 °C – 66 °C	78 °C	92 °C	Inspect cooling circuit at warning
Operating Pressure	205.0 – 215.0 bar	185.0 bar (Low Pressure)	175.0 bar (Low-Low Press)	Halt cycle at low-pressure alarm
Motor Current (Full Load)	115 A – 142 A	158 A	172 A	Check proportional valve load

2.2 Hazardous Energy Isolation Prior to performing any mechanical or hydraulic intervention, technicians shall execute complete isolation of the following hazardous energy sources in accordance with Vindhya Precision Works Lockout/Tagout (LOTO) Procedure EHS-LOTO-01: 1. Hydraulic Energy: Isolate main pump circuit breaker, close manual isolation valve V-01, and bleed accumulator pressure to < 0.5 bar via drain valve DV-01. 2. Electrical Energy: Open and lock out main 415V supply isolator Q1 at Motor Control Centre MCC-HYD-02. 3. Gravity: Lower the 400-tonne main slide to the mechanical dead-stops and engage the hydraulic drop-pin safety locks (Pins LP-01 and LP-02).

3. PREVENTIVE MAINTENANCE SCHEDULE

3.1 Maintenance Responsibility * Responsible Trade: Hydraulics * Supporting Trade: Mechanical * Base Preventive Maintenance Interval: Exactly 750 running hours (± 12 hours tolerance).

3.2 Running-Hour Task Matrix * 750 Hours (PM-1): * Inspect proportional valves for spool stiction. * Grease main slide bronze guideways with VPW-approved synthetic grease (15 strokes per nipple). * Check electrical terminations at contactor VPW-P-00042. * Clean return-line filter housing. * 1,500 Hours (PM-2): * Perform PM-1 tasks. * Replace return-line and pressure-line filtration elements. * Inspect drive belt tension; replace Economy belt VPW-P-00037 if deflection exceeds 8 mm under 3.5 kg load. * Calibrate pressure transducers against reference gauge. * 4,500 Hours (PM-3): * Perform PM-2 tasks. * Overhaul main cylinder dynamic seals using Deep seal VPW-P-00002 and Inside seal VPW-P-00005. * Inspect and replace secondary sealing elements (Almost seal VPW-P-00032). * Inspect Camera bearing VPW-P-00013 and Account bearing VPW-P-00024 for radial play exceeding 0.05 mm; replace if necessary. * Verify alignment of Left sensor VPW-P-00009.

4. FAULT-CODE TABLE AND TROUBLESHOOTING

Table 4.1: HYD-PRESS Fault Conditions, Causes, and Corrective Actions
Fault ID	Condition Description	Probable Cause	Corrective Action
F-HYD-01	Main Pump Cavitation Noise	Restricted suction strainer or degraded oil viscosity at start-up.	Clean suction strainer; verify hydraulic oil temperature is above 25 °C before starting pump.
F-HYD-02	Slide Creep at Top Dead Center	Internal bypass across main piston seals (Deep seal VPW-P-00002 / Inside seal VPW-P-00005).	Isolate energy sources, perform LOTO, and replace cylinder seal kits.
F-HYD-03	Press Fails to Build Tonnage	Proportional relief valve sticking or pilot pressure failure below 28.0 bar.	Clean pilot stage valve spool; verify proportional amplifier card output current (target 400-800 mA).
F-HYD-04	Excessive Vibration Alarm (5.6 mm/s)	Misalignment of pump-motor coupling or loose mounting bolts on base plate.	Check coupling alignment with laser tool (tolerance < 0.05 mm); torque motor foot bolts to 210 Nm.
F-HYD-05	High Surface Temperature (78 °C)	Heat exchanger fouling or insufficient cooling water flow (< 45 L/min).	Backwash shell-and-tube heat exchanger; verify cooling water inlet temperature does not exceed 25 °C.
F-HYD-06	Contactor Chatter / Failure	Low control voltage or pitted main contacts on Notice contactor VPW-P-00042.	Measure control circuit voltage (must be 110V AC $\pm 5\%$). If voltage is stable, replace contactor VPW-P-00042.

5. CONSUMABLE AND SPARE PARTS LIST Technicians shall only use inventory items matching the exact part numbers listed in Table 5.1. Substitution of unlisted parts is strictly prohibited.

Table 5.1: Approved Spare Parts and Consumables Inventory

Part Number	Description	Application
VPW-P-00002	Deep seal	Main hydraulic cylinder primary dynamic sealing element
VPW-P-00005	Inside seal	Main hydraulic cylinder secondary wiper and rod seal
VPW-P-00009	Left sensor	Slide position linear encoder / proximity sensor assembly
VPW-P-00013	Camera bearing	Optical alignment guide bearing on upper crown assembly
VPW-P-00024	Account bearing	Main toggle pivot thrust bearing assembly
VPW-P-00032	Almost seal	Auxiliary manifold block static O-ring kit
VPW-P-00037	Economy belt	Main hydraulic pump auxiliary auxiliary drive belt
VPW-P-00042	Notice contactor	Main motor control panel primary power switching contactor

6. TORQUE SPECIFICATIONS All threaded fasteners associated with the HYD-PRESS asset class shall be torqued to the values specified in Table 6.1 using a calibrated torque wrench. Threads shall be lightly coated with anti-seize compound unless specified otherwise.

Table 6.1: Fastener Torque Matrix

Fastener Location	Thread Size	Target Torque	Remarks
Main Cylinder Flange Bolts	M30 (Grade 12.9)	1,450 Nm	Star pattern tightening in 3 stages (500 Nm, 1000 Nm, 1450 Nm)
Pump-Motor Mounting Foot	M16 (Grade 8.8)	210 Nm	Cross-pattern torque
Hydraulic Manifold Tie-Bolts	M12 (Grade 10.9)	85 Nm	Sequential torque pass
Base Frame Anchor Bolts	M24 (Grade 8.8)	650 Nm	Verify with hydraulic tensioner every 4,500 hours

7. SAFETY-CRITICAL CORRECTIVE ACTIONS

7.1 Mandatory LOTO and Permit Requirements The following corrective actions involve high-energy release risks or heavy lifting overhead. Technicians shall not commence work without executing a formal Confined Space / Mechanical LOTO Permit countersigned by the Area Safety Engineer: 1. Replacement of Main Cylinder Seals (Deep seal VPW-P-00002 and Inside seal VPW-P-00005): Requires mechanical pinning of the 400-tonne slide, absolute pressure zeroing of accumulators, and overhead crane lifting authorization. 2. Replacement of Account bearing VPW-P-00024: Requires hydraulic cylinder shoring and structural load-transfer rigging to prevent accidental slide collapse. 3. Replacement of Notice contactor VPW-P-00042: Requires electrical LOTO at MCC-HYD-02 with mandatory voltage verification using a CAT IV multimeter confirming 0.0V across all terminals prior to terminal touching. 4. Intervention on High-Pressure Hydraulic Lines (> 185 bar): Requires full depressurization, zero-energy verification via gauge PT-01, and thermal cool-down until surface oil temperature drops below 40 °C.